Abhishek Gond

Software Engineer | AI & Full-Stack Developer

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Education

- **Institute of Engineering and Technology** — Lucknow, India
  B.Tech in Electronics and Communication Engineering; CGPA: 7.85/10 — *2023 – 2027 (Expected)*
- **Class XII:** 83.6% | **Class X:** 83.8%

Experience

- **Software Engineer Fellowship** — *Jul 2026*
  Panscience Innovations | Node.js, Express 5, TypeScript, Next.js, MongoDB, Pinecone, RAG, LLM
  - Built an Express 5/TypeScript backend for multimodal clinical PDF extraction, verifying each value against source text and handling LLM failures via fallback and circuit breaker.
  - Built Pinecone RAG (local BGE-small embeddings, JWT tenant isolation) on MedlinePlus/MeSH, ACC/AHA, WHO guidelines.
  - Blocked hallucinated citations via in-code validation and `insufficient_evidence` handling; 270 backend + 10 frontend tests.
- **Full-Stack Developer Intern** — *Feb 2026 – Apr 2026*
  Durga Foundation | Next.js, Node.js, Prisma, Redis, MySQL
  - Migrated the NGO portal to Next.js and built the "Youth Corner" full-stack platform (Node.js, Express, MySQL).
  - Architected an 11-client multi-schema Prisma setup; integrated Redis caching and Razorpay donation processing.

Projects

- **GridSense: AI Solar Panel Monitoring Platform** | *Next.js, Node.js, PostgreSQL, Socket.IO, Prisma, pgvector* — Live Demo
  - Streamed simulated panel telemetry (voltage, current, temperature, irradiance) every **2 s** via **Socket.IO** to per-user rooms, batching 30 readings/panel into one Prisma write per minute.
  - Designed a **LangGraph copilot** where Groq Llama 3.1 8B calls Zod-validated Prisma tools instead of generating SQL to explain soiling, shading and overheating faults, with pgvector retrieval over 384-d MiniLM embeddings.
  - Implemented rotating **httpOnly refresh-token sessions** (15-min JWTs) and Razorpay HMAC-SHA256 verification with an atomic Prisma transaction across payment, subscription and user records.
- **Algorithmic RAG Engine (Framework-Free)** | *Python, NumPy, FastAPI, Next.js, Gemini* — Live Demo
  - Built a framework-free RAG pipeline in Python: O(N) sliding-window chunker (200-word windows, 50-word overlap) feeding local CPU MiniLM embeddings, with no LangChain or LlamaIndex.
  - Coded exact top-k retrieval using vectorized NumPy cosine similarity on an in-memory M×384 float32 matrix; model pre-warmed.
  - Designed FastAPI ingest/query endpoints separating retrieval from generation; Gemini answers with source citations under a strict grounding prompt, with model fallback and per-stage latency reporting.
- **Afterword: AI Video Intelligence Platform** | *Python, LangChain, Whisper, Mistral, ChromaDB* — Live Demo
  - Built a yt-dlp/ffmpeg audio pipeline (16 kHz mono, 10-min chunks) using local Whisper for English and Sarvam AI for one-step Hinglish-to-English transcription (25 s chunks).
  - Implemented Mistral 8B map-reduce summarization (3000-char chunks, 200 overlap) with 3 LCEL chains extracting action items with owners and deadlines, decisions and open questions.
  - Added a grounded MiniLM/ChromaDB RAG chat (top-4) that flags questions the video doesn't cover instead of hallucinating.

Technical Skills

- **Languages:** Python, C++, JavaScript, TypeScript, SQL
- **Frontend:** HTML/CSS, React.js, Next.js, Tailwind CSS, Framer Motion, GSAP, Lenis
- **Backend:** Node.js, Express.js, FastAPI, REST APIs
- **AI/ML & GenAI:** Machine Learning, LLMs, GenAI, RAG, LangChain, LangGraph, AI Agents, Vector Search
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Prisma, pgvector, ChromaDB, Pinecone
- **Tools & Core CS:** Git, Docker, CI/CD, pytest, OOP, DBMS, Operating Systems, Computer Networks

Hackathons & Competitions

- **Hack Devengers 1.0 – Top 30 / 4.2K+ Participants – (CodeRAG Vector)** — Live Demo | *Aug 2026*
  - Engineered a FastAPI repository-audit pipeline ingesting local paths or GitHub zipballs, with directory pruning, extension whitelisting and 15-file/300-line caps to bound LLM token usage.
  - Grounded audits in 17 OOP, DBMS, concurrency and network-security rules (MiniLM, ChromaDB), retrieving the top-5 per file.
  - Generated per-file 0–100 scores with rule-cited violations, fix plans and refactored code via Gemini JSON mode, validated by Pydantic with heuristic fallbacks and pytest coverage.
- **Smart India Hackathon (SIH) Internal Round – 3rd Rank** — *Sep 2026*
  - Architected **AGNI.AI**, a disaster response system fusing NASA FIRMS satellite telemetry, PostGIS mapping, and multimodal AI (VLM + XGBoost) to classify industrial thermal anomalies and trigger alerts.

Achievements & Certifications

- Data Structures & Algorithms: Solved 450+ problems on LeetCode and GFG, focusing on time/space complexity optimization.
- **HackerRank Certified – SQL (Advanced)** — View Certificate | *Oct 2026*
- **HackerRank Certified – REST API (Intermediate)** — View Certificate | *Oct 2026*
- **HackerRank Certified – Problem Solving (Intermediate)** — View Certificate | *Oct 2026*

Volunteer Experience

- **Core Member, Training & Placement Cell (TPC)** — *Sep 2025 – Present*
  Student liaison between hiring teams and candidates; guided peers through the campus recruitment process.